

Exploring High-Pass and Low-Pass Dynamics in RC Circuits

INTRODUCTION

Resistance capacitor circuits serve as key components in timing, filtering, and signal shaping. In this lab, we analyze the RC time constant and study how the circuit’s output changes with frequency.

THEORY

(~) The voltage ratio for a low-pass filter is given by:

wh $\left| \frac{V_{out}}{V_{in}} \right| = \frac{1}{\sqrt{1 + (2\pi f\tau)^2}}$ and τ is the time constant.

(~) The time constant is defined as: $\tau = R \times C$.

(~) At low frequencies, the output voltage is $\approx$ equal to the input.

(~) At higher frequencies, the capacitor’s impedance decreases, altering the output.

(~) The observed change in V_{out} / V_{in} versus frequency indicates whether the circuit behaves as a low-pass or high-pass filter.

METHODS

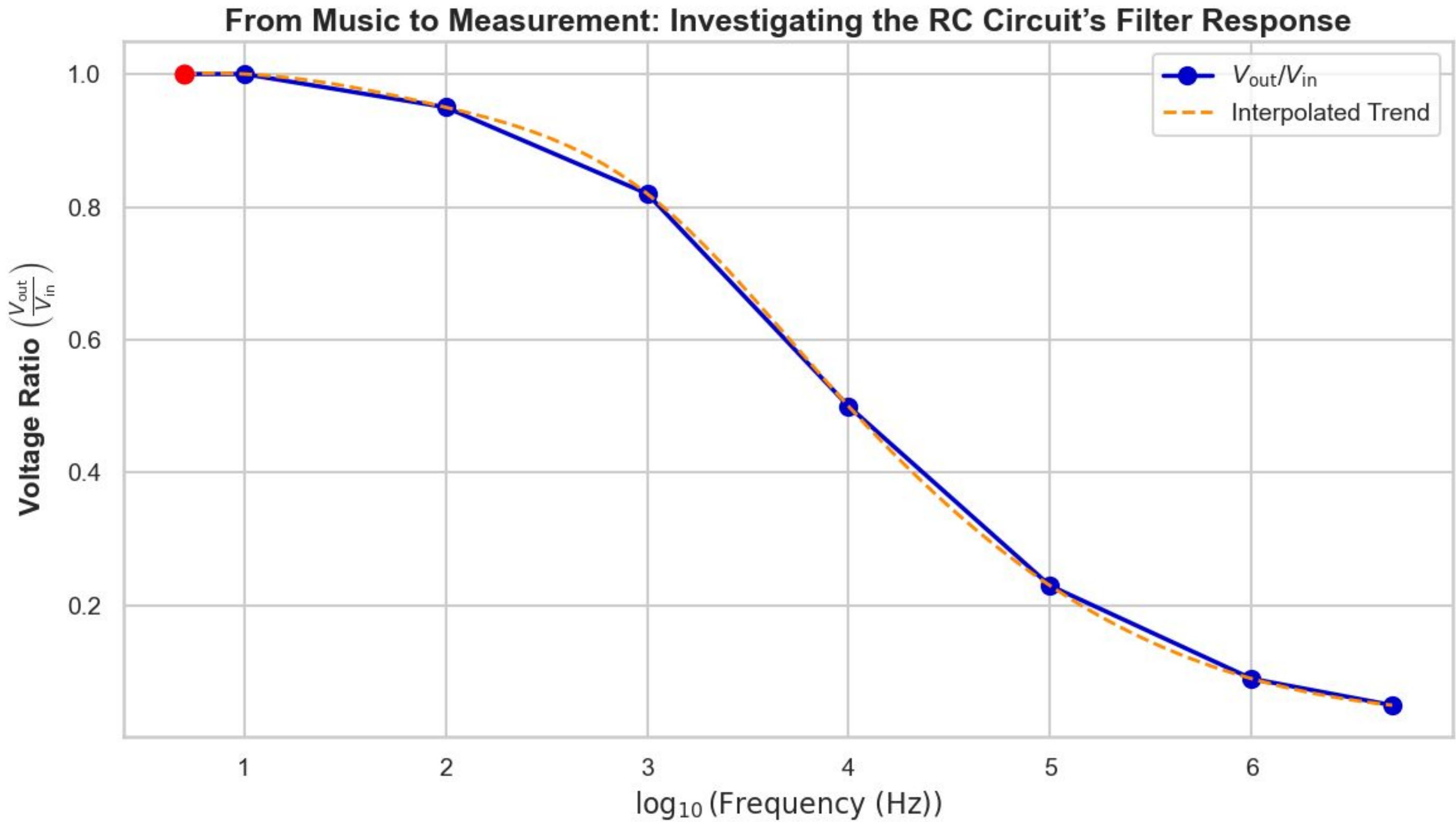
A function generator, oscilloscope, and RC components were used to build the circuit. The theoretical time constant ($\tau = R \times C$) was calculated and compared with oscilloscope measurements. The frequency response was determined by plotting $V_{ou} \square / V_i \square$ versus frequency to identify the threshold behavior.

Dominant Source of Uncertainty

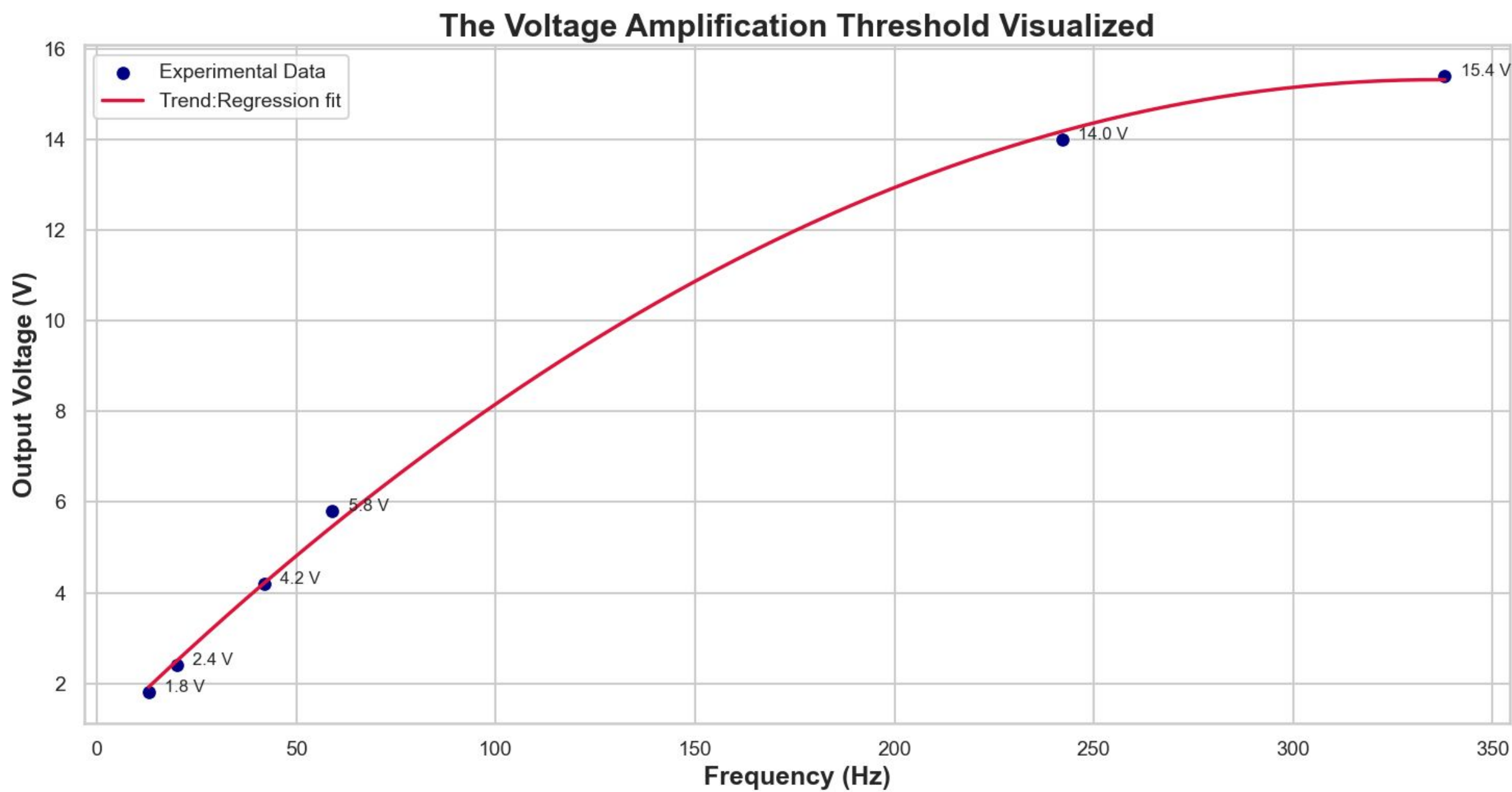
The dominant source of uncertainty in the time constant measurement is the tolerances of the resistor and capacitor. Using uncertainty propagation, the relative uncertainty in the time constant is given by:

$$\frac{\Delta \tau}{\tau} = \sqrt{\left(\frac{\Delta R}{R}\right)^2 + \left(\frac{\Delta C}{C}\right)^2}$$

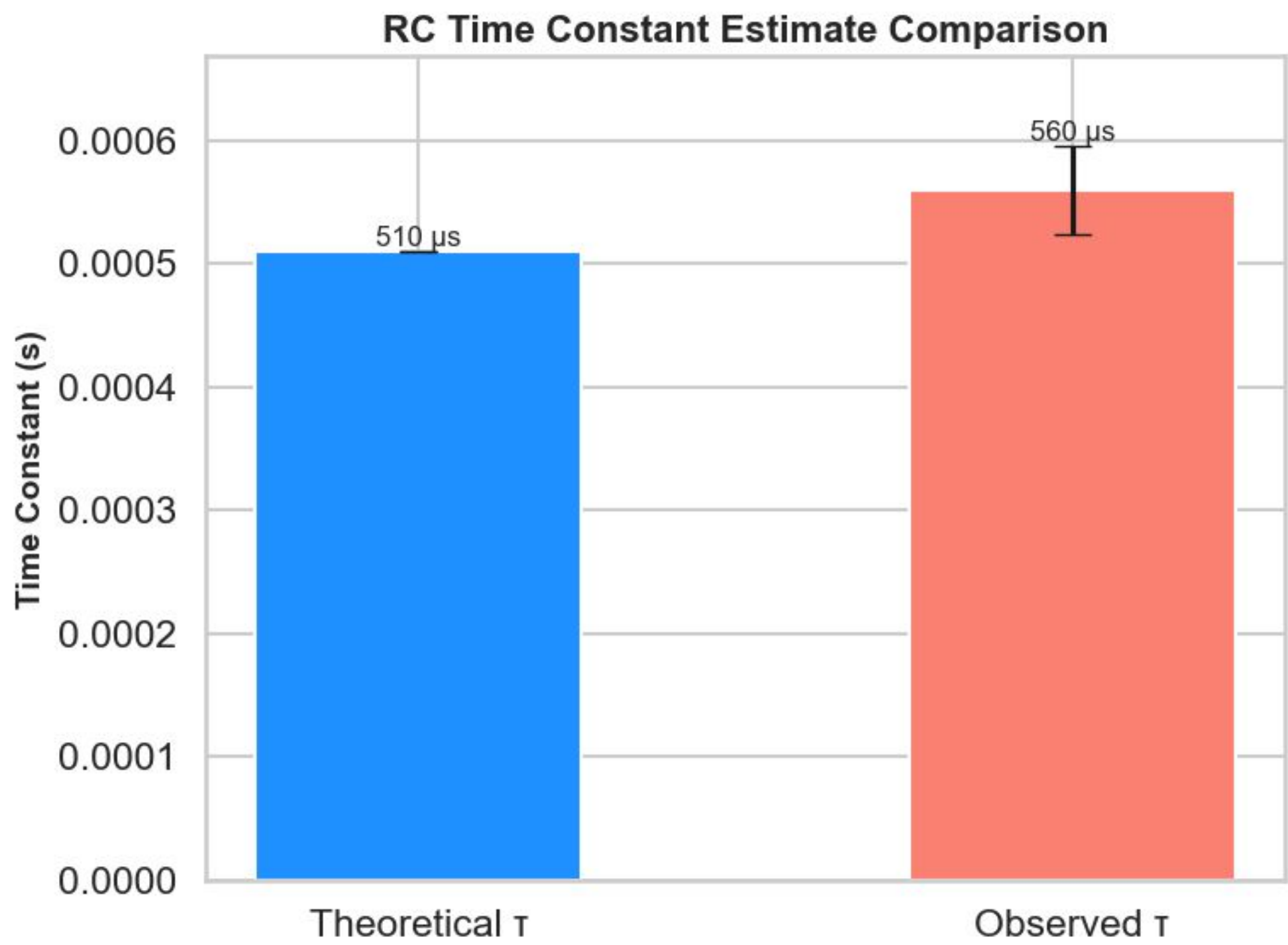
Findings and Results



The theoretical time constant (510 μ s) +/- 36 closely matches the observed value (560 μ s), validating the RC circuit model. Additionally, the frequency response plot shows that the voltage ratio V_{out} / V_{in} increases from about 0.1 at low frequencies to approximately 0.87 at higher frequencies, indicating that the circuit acts as a high-pass filter.



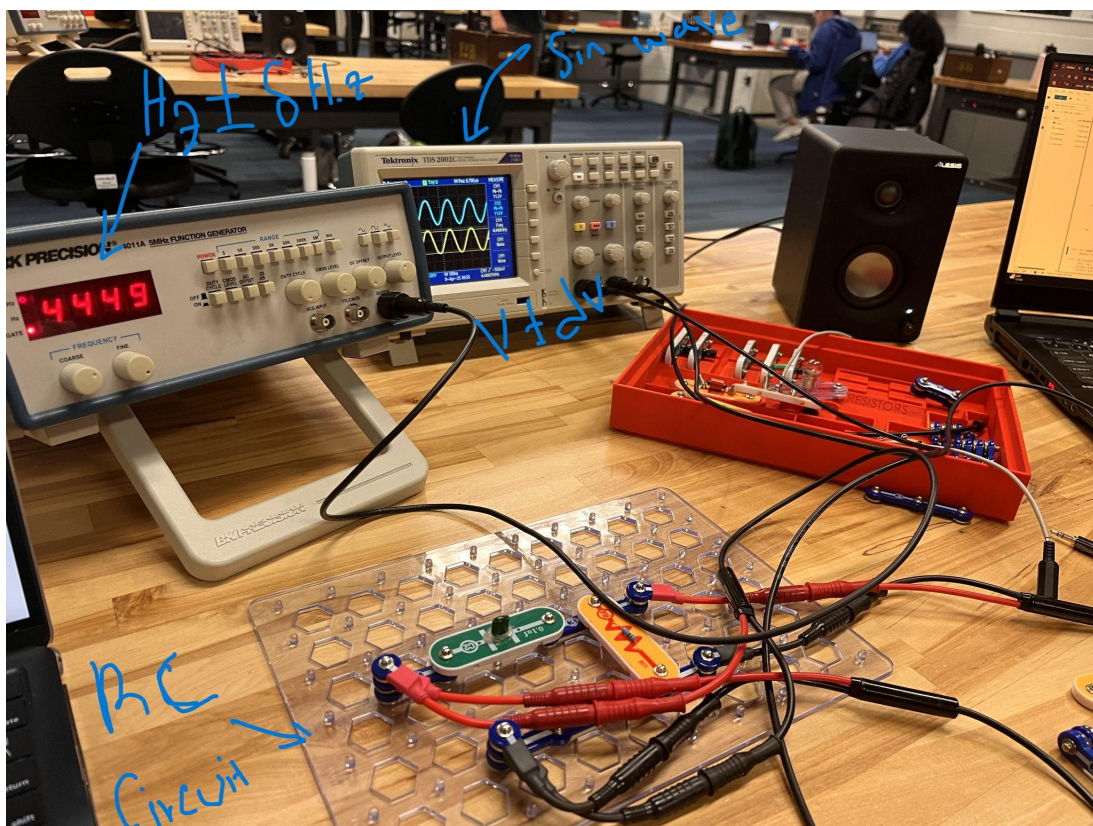
Comparison To Accepted Value



Experimental Reflection

The experiment demonstrated that the RC circuit model not only accurately predicts circuit behavior but also distinguishes between high-pass and low-pass characteristics. The high circuit, which allows higher frequency signals to pass, can be likened to the crisp, clear treble found in modern electronic music, while the low circuit, which attenuates high frequencies, mirrors the deep, resonant bass tones featured in classical and jazz compositions.

Apparatus



Analysis PDF



SCAN ME